Experiment 17: VoteOnline – E-Voting Platform

1. Aim

To design and develop a user-friendly e-voting website named VoteOnline using HTML, CSS, and JavaScript, enabling administrators to manage elections, candidates, and voter lists, while allowing voters to log in using their Voter ID, verify through OTP, select a candidate, submit their vote, and view voting history and election results.

2. Procedure

1. Create the project folder named VoteOnline and create the files index.html, style.css, and app.js.
2. Design the user interface using HTML and style it using CSS to provide a responsive layout.
3. Implement separate login interfaces for administrators and voters.
4. Add administrator functionalities to create elections, add candidates, allocate candidates to elections, manage voter lists, import voter data using CSV, and view election results.
5. Implement voter authentication using Voter ID and simulated OTP verification.
6. Display the candidates available for the voter's assigned ward.
7. Allow voters to select a candidate and verify their choice using a simulated voting OTP.
8. Store submitted votes and voter history in the browser's localStorage.
9. Generate a confirmation receipt after successful vote submission and prevent duplicate voting within the same election.
10. Display election progress and candidate-wise vote counts.
11. Execute index.html in a web browser and test the administrator and voter functionalities.

3. Code Snippet

A. HTML – Login interface (index.html)

```
<h2>VoteOnline</h2>

<form id="voterLoginForm">

  <label for="voterIdInput">Voter ID</label>

  <input id="voterIdInput"

      placeholder="Enter Voter ID"

      required>

  <button type="submit">Login</button>

</form>
```

```
<div id="loginMessage"></div>
```

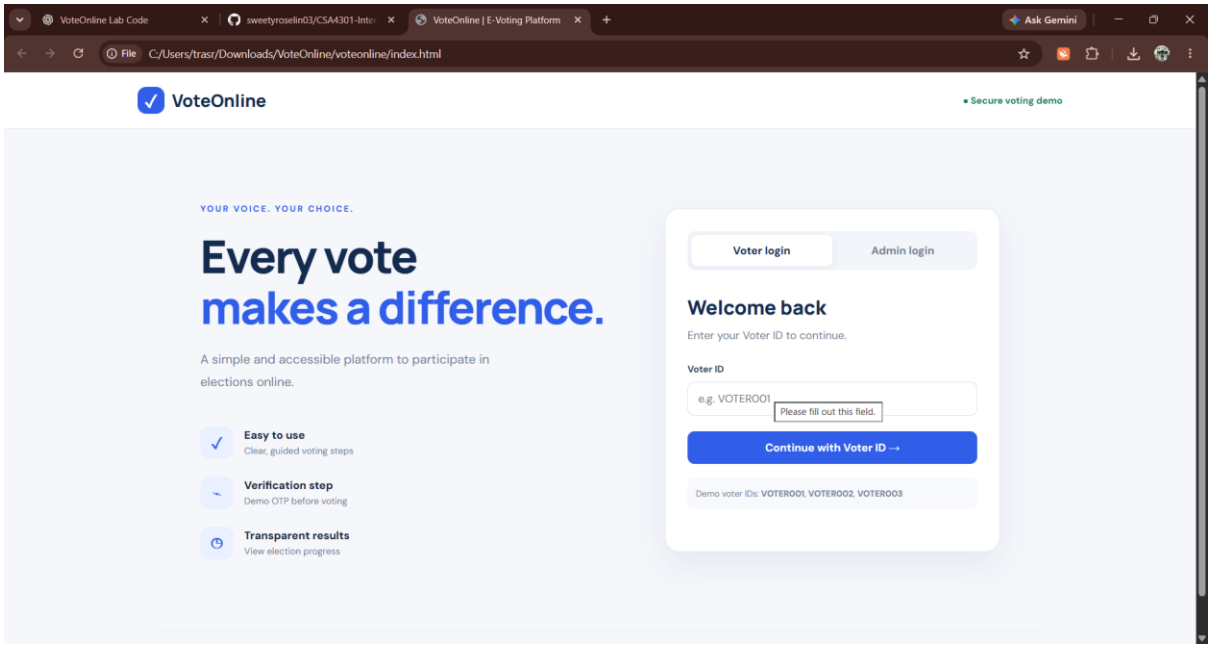
B. CSS – Styling (style.css)

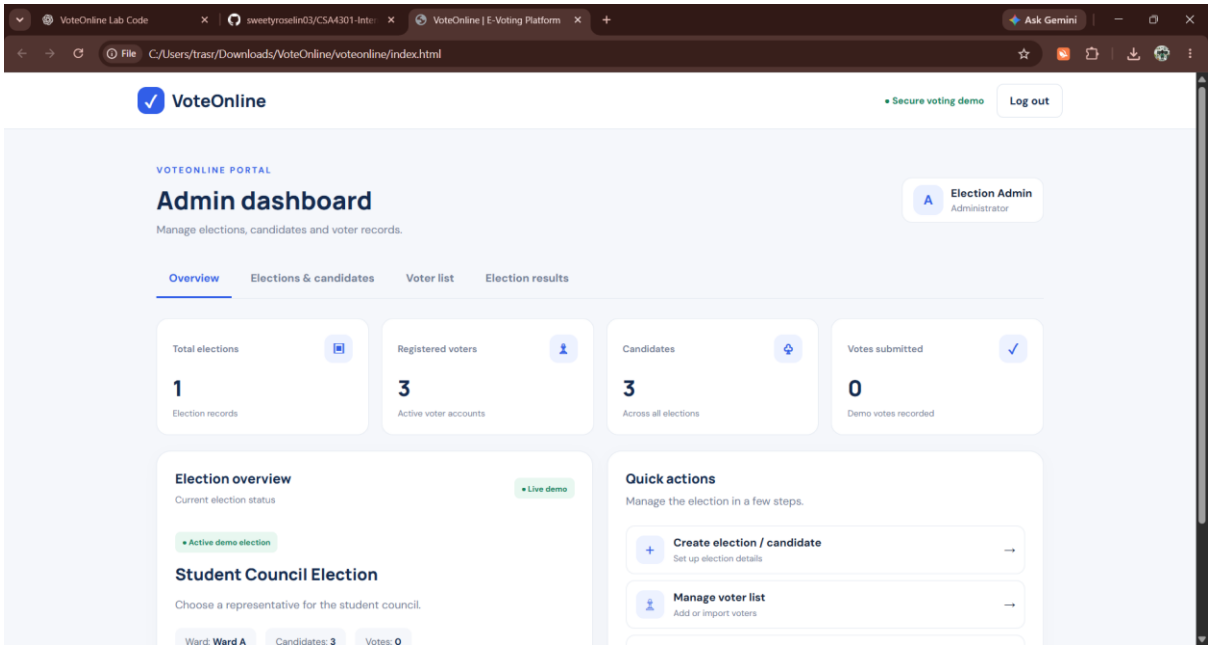
```
body {
    font-family: Arial, sans-serif;
    background: #f5f7fb;
    margin: 0;
    padding: 30px;
}
.login-card {
    max-width: 400px;
    margin: 50px auto;
    padding: 25px;
    background: white;
    border-radius: 12px;
    box-shadow: 0 4px 15px #0001;
}
input, button {
    width: 100%;
    padding: 12px;
    margin-top: 10px;
    box-sizing: border-box;
}
button {
    background: #315ee8;
    color: white;
    border: none;
    border-radius: 6px;
    cursor: pointer;
}
```

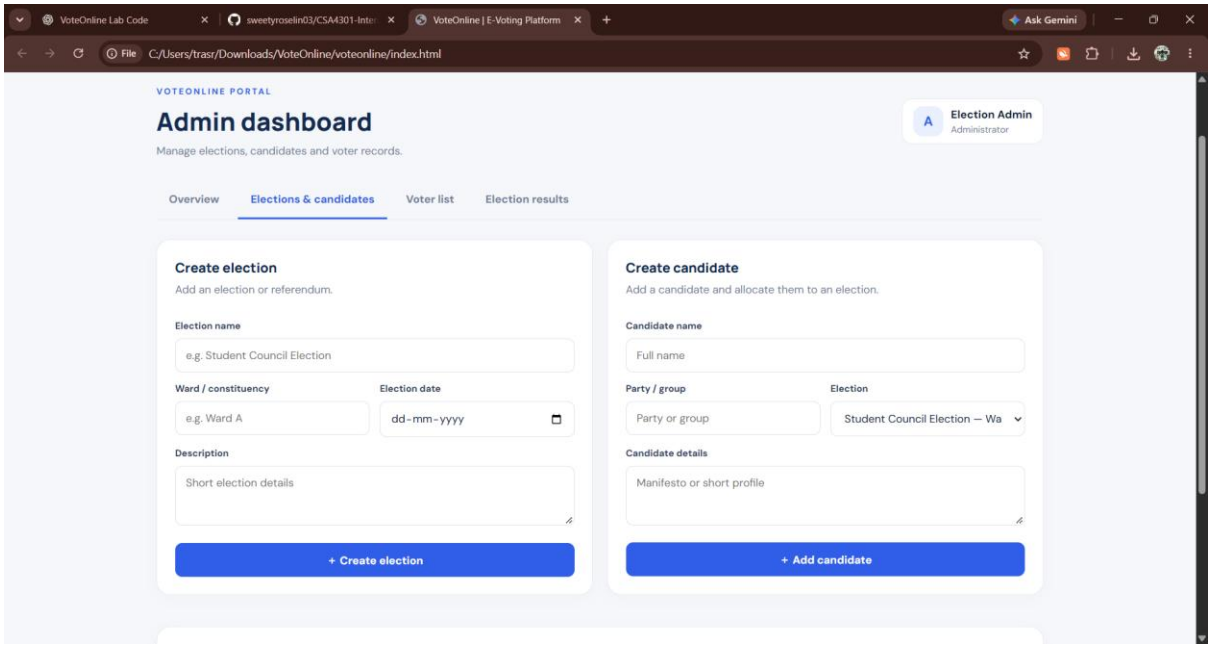
C. JavaScript – Voter authentication (app.js)

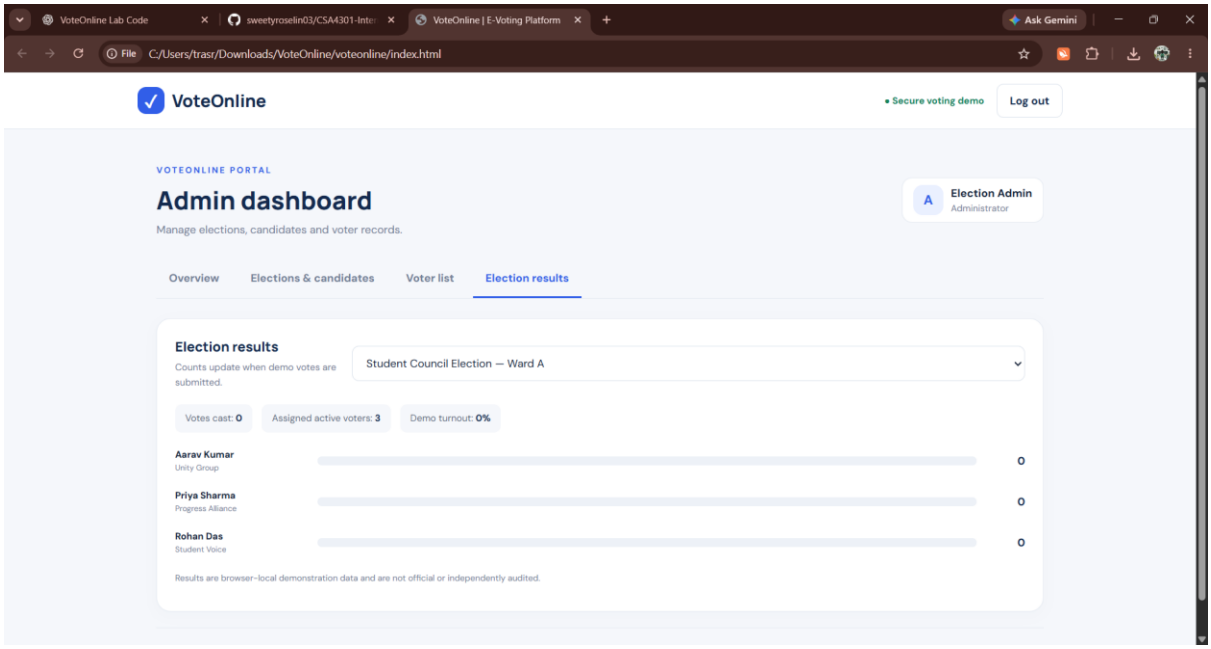
```
const demoOTP = "123456";
document.getElementById("voterLoginForm")
    .addEventListener("submit", function (event) {
        event.preventDefault();
        const voterId = document
            .getElementById("voterIdInput")
            .value.trim().toUpperCase();
        const voters = ["VOTER001", "VOTER002", "VOTER003"];
        if (voters.includes(voterId)) {
            alert("Your demo OTP is: " + demoOTP);
            const otp = prompt("Enter OTP:");
            if (otp === demoOTP) {
                document.getElementById("loginMessage")
                    .textContent = "Login successful!";
            } else {
                document.getElementById("loginMessage")
                    .textContent = "Invalid OTP!";
            }
        } else {
            document.getElementById("loginMessage")
                .textContent = "Invalid Voter ID!";
        }
    });
```

4. Output


VoteOnline Lab Code
sweetyroselin03/CSA4301-Inter
VoteOnline | E-Voting Platform
Ask Gemini
File C:/Users/trasr/Downloads/VoteOnline/voteonline/index.html
VoteOnline
• Secure voting demo
YOUR VOICE. YOUR CHOICE.
Every vote
makes a difference.
A simple and accessible platform to participate in elections online.
Easy to use
Clear, guided voting steps
Verification step
Demo OTP before voting
Transparent results
View election progress
Voter login
Admin login
Welcome back
Enter your Voter ID to continue.
Voter ID
e.g. VOTER001
Please fill out this field.
Continue with Voter ID →
Demo voter IDs: VOTER001, VOTER002, VOTER003


VoteOnline Lab Code
sweetyroselin03/CSA4301-Inter
VoteOnline | E-Voting Platform
Ask Gemini
File C:/Users/trasr/Downloads/VoteOnline/voteonline/index.html
VoteOnline
• Secure voting demo
Log out
VOTEONLINE PORTAL
Admin dashboard
Manage elections, candidates and voter records.
A
Election Admin
Administrator
Overview
Elections & candidates
Voter list
Election results
Total elections
1
Election records
Registered voters
3
Active voter accounts
Candidates
3
Across all elections
Votes submitted
0
Demo votes recorded
Election overview
Current election status
• Live demo
• Active demo election
Student Council Election
Choose a representative for the student council.
Ward: Ward A
Candidates: 3
Votes: 0
Quick actions
Manage the election in a few steps.
Create election / candidate
Set up election details
Manage voter list
Add or import voters





5. Result

Thus, the VoteOnline e-voting website was designed and developed successfully using HTML, CSS, and JavaScript. The application demonstrates administrator election management, voter authentication, OTP verification, candidate selection, vote submission, confirmation receipts, voting history, and election result visualization.